CSE 6093: Research Methodologies and Technical Writing

Prof. Dr. Khondaker Abdullah Al Mamun Date: 10-July-2026

Key Outcomes

The professor introduced an MSc-level research methodology course at UIU, emphasizing a **project-based, skill-driven approach** over traditional exam-focused learning. Students are expected to independently complete a full research cycle — from problem identification to journal-quality writing. Two teaching assistants (Fiona and shohag) were selected to support coordination. The course runs in **hybrid mode** with both online and physical sessions.

Decisions Made

- **Assessment structure:** Assignments 25 marks | Mid-term project presentation 30 marks | Final presentation 40 marks — no traditional written exams. *1*
- **Individual projects only:** Each student works solo through the full research pipeline to develop complete researcher competency. *2*
- **Teaching assistants selected:** One male (shohag) and one female (Fiona) student appointed to bridge communication gaps between students and instructor. *3*
- **Recommended textbook:** *Principles of Research in Behavioral Science* — lecture notes provided if book is unavailable. *2*

Course Philosophy & Approach

- **Learning by doing** over memorization; focus on practical skill development. *4*
- Research pipeline each student must complete: identify problem → analyze → decompose → literature review → propose innovative solution → pilot/POC → present → write full research article. *56*
- **Professor's philosophy:** "Solve the problem, develop a product or service — the journal is a byproduct." *7*
- Mid-term and final are **project showcases**, not written tests. *6*

Mindset Framework Emphasized

- **5WH framework** (Who, what, When, Where, why, How) embedded as core investigative habit for all research and professional work. *89*
- **Curious mindset** cited as the only true prerequisite — no specific academic background required. *1011*
- **Entrepreneurial mindset** stressed: ability to define, lead, and develop new processes/solutions; contrasted with follower mentality. *912*
- Key traits required: **tenacity, intuition, rationalism, and critical thinking** — especially important in the AI era. *13*

Student Introductions — Cohort Snapshot

Diverse backgrounds represented, including:

- **CSC/Engineering:** Multiple students from IUB, UIUTK, MIST, BUP with backgrounds in CSE, petroleum engineering, and mathematics. *71415*
- **Industry professionals:** Students currently working as software engineers, corporate trainers, civil aviation staff, and petroleum corporation employees. *1617*
- **Research-active:** At least one student noted a prior Q1 journal publication as first author. *18*

Pending Confirmation

- Exact schedule split between online and physical classes to be communicated (professor traveling to Bangladesh periodically). *19*

- Students without textbook access to receive lecture notes — delivery method not specified. *2*

Action Items

- **All students:** Begin thinking about individual research problem selection for project work.
- **Fiona & shohag (TAs):** Ensure no information gaps between instructor and students; communicate all course updates. *19*
- **Students interested in online attendance:** Must join recorded sessions; professor prefers recorded online classes over live passive attendance. *19*

Connect with Shohag

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